

Mariam Ossama

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SUMMARY

Senior Computer Engineering student with hands-on experience in machine learning, deep learning, and computer vision. Strong background in Python, SQL, and Power BI, with applied knowledge of neural networks, CNNs, transfer learning, and advanced computer vision tasks including image classification, object detection, segmentation, and face recognition. Experienced in data engineering, ETL workflows, and data visualization through academic projects and internships, delivering scalable and insight-driven solutions.

EDUCATION

Bachelors of Computer & Systems Engineering, Helwan university

2021 – 2026

Embedded Systems Minor

Overall grade of excellence - 89.3%

PROFESSIONAL EXPERIENCE

Ejada, Data and analytics engineering intern

07/2025 – 08/2025

Gained hands-on experience in data engineering tasks including ETL processes and dataset preparation where I Built a data warehouse to support structured reporting and analytics, followed by data visualization and reporting using Power BI

NTI, Deep learning for computer vision

08/2025

Gained hands-on skills in neural networks, CNNs, transfer learning, modern architectures and Downstream tasks including classification, detection, segmentation, and face recognition

DEPI, IBM Data Scientist

11/2024 – 05/2025

Completed a comprehensive program covering Python, SQL, data analysis, visualization, and machine learning and model deployment. Gained hands-on experience with real-world datasets, building predictive models along with a graduation project at the end.

ITI, Embedded Software Intern

07/2024 – 08/2025

Acquired hands-on experience in embedded C programming for AVR microcontrollers. Implemented projects involving timers, interrupts, communication protocols and peripheral interfacing, with a focus on designing reliable and efficient embedded systems.

Banque du caire, IT intern

08/2023

Acquired hands-on exposure to banking IT infrastructure, system components, operational processes, and collaborative workflows across technical teams.

PROJECTS

VideoRag

Built a custom, low-level Retrieval-Augmented Generation (RAG) lito enable semantic Q&A over videos and documents. The system extracts transcripts and visual text, performs vector-based retrieval, and generates context-aware responses from relevant video segments.

ML-Based FTP Intrusion Detection System

Built a machine learning-based IDS to detect FTP brute-force attacks using real network traffic. Trained and evaluated models on 720K+ packets, achieving ~99% accuracy and ~98% recall using a Random Forest classifier.

ETA : E-learning Talking Avatar, Helwan cyber arena Hackathon'25

Applied computer vision skills to implement a face-to-avatar system for lip-sync and facial animation. Successfully ran and optimized the model locally to generate animated avatars that deliver educational content, enhancing engagement and interactivity in e-learning platforms.

Garbage Classifier App

Built a desktop application using Python (GUI) integrated with YOLOv8 for object detection. Applied transfer learning with MobileNet to classify and sort waste types efficiently

End-to-End Business Intelligence Solution, Ejada

Designed and implemented a data warehouse using dimensional modeling, ETL pipelines, and SCD to centralize and optimize sales data. Built interactive Power BI dashboards (Sales, Customer, Product, Orders) with KPIs, drill throughs, and trend analysis, enabling efficient reporting and data-driven decision-making

Heart Disease Prediction App, DEPI

Built a machine learning application using Python and Random Forest to predict the likelihood of heart disease. Performed exploratory data analysis (EDA) and data visualization to identify key health patterns and improve model accuracy

Grocery Store App

Developed an interactive Flutter application for grocery shop automation, featuring product browsing, cart management, and order handling to streamline shopping operations.

Smart Home Automation, ITI

Built a microcontroller-based system using ATmega32 in embedded C to automate and control devices using timers, PWM, and interrupts, with added security and real-time features.

SKILLS

Programming Languages

Python, C/C++, Java, Embedded C, VHDL, Dart, JavaScript, HTML, CSS

Machine Learning & Artificial Intelligence

Supervised Learning, Unsupervised Learning, Model Training, Model Evaluation, Feature Engineering, Predictive Modeling, Statistical Modeling, Neural Networks, RNN basics, CNNs, Transfer Learning.

Computer Vision

Image Processing, Video Processing, Image Classification, Object Detection, Image Segmentation, Face Recognition.

Frameworks & Libraries

PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Flutter

Data Engineering & Analytics

ETL Pipelines, Data Warehousing, Data Integration, Dimensional Modeling, Slowly Changing Dimensions (SCD), SQL, Power BI, Data Visualization

Embedded Systems & Hardware

Microcontrollers (ARM, AVR), Firmware Development, RTOS Fundamentals, FPGA Design (VHDL), Digital Design, Embedded Software Development

Tools & Platforms

Git, GitHub, Linux, Bash, Proteus, Wireshark.

Soft Skills

Committed, Fast Learner, Critical Thinking, Team Collaboration, Technical Communication, Adaptability, Time Management

AWARDS

1st place AI-Battle Hackathon'25,

15/09/2025

Helwan Cyber Arena sponsored by Cysheild and EG-CERT

LANGUAGES

• English

• German

• Arabic